Om Singhal

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Education

- **Indian Institute of Information Technology Jabalpur** *Aug 2023 - May 2027 (Expected)*
  Bachelor of Technology - Computer Science Engineering Jabalpur, India

Achievements

- **LeetCode: omsinghal852**
  - Reached **Guardian** status (2198 rating, top 1% globally) with 700+ problems solved.
  - Ranked 269 in Weekly Contest 495 and 229 in Biweekly Contest 191 (∼20K participants each).
- **CodeForces: omsinghal852**
  - Reached **Expert** status (1619 rating) with 600+ problems solved on Codeforces.
  - Ranked 358 globally among 30K+ participants in Codeforces Round 1103 by solving 6 problems.
- **ICPC Amritapuri 2025–26:** Qualified as an **ICPC Regionalist** (Rank **632**). Collaborated in a 3-person team to analyze ambiguous problem sets, peer-review logic, and design optimal solutions under strict time constraints.
- **CodeRumble 4.0 (IIIT DM Jabalpur):** Won **1st Prize** out of **100+ teams** in a multi-college, offline ICPC-style contest; solved 4 problems, the most of any team. View Certificate

Projects

URL Shortener | *Tools: FastAPI, React, Vite, Tailwind CSS, PostgreSQL, Redis, Docker, Nginx, Locust* *Link* | *Github*

- Built a URL shortener backend (FastAPI, PostgreSQL/`asyncpg`, Redis) with a Redis cache-first read path cutting redirect latency ∼**3.7x** and a write-buffering layer batching click events into idempotent, crash-safe PostgreSQL flushes every 60s; deployed **4 replicas** behind Nginx with least-connections load balancing.
- Diagnosed a Postgres connection-pool exhaustion bug via Locust load testing (aggregate pool size of 240 exceeded the 100-connection limit, causing **45.6% request failures**); fixed by tuning per-replica pool sizing, reducing failures to **0%** across 100-300 concurrent users.
- Characterized system capacity under load: ∼**3,000 RPS** at 31ms avg latency (100 users), with backend CPU saturation (∼95%) capping throughput at ∼**1,600 RPS** by 200+ users, showing graceful degradation instead of failure under overload.

RAG_ASK | *Tools: Python, LangChain, Gemini API, FAISS, HuggingFace, Streamlit* *Link* | *Github*

- Developed a Retrieval-Augmented Generation chatbot for IIITDMJ institutional queries, using LangChain to orchestrate retrieval and Google's Gemini API (`gemini-flash-lite-latest`) for response generation.
- Embedded documents using the HuggingFace `all-MiniLM-L6-v2` model and indexed them in a local FAISS vector store for low-latency semantic search, grounding responses in retrieved context to reduce hallucinations.

Leadership & Positions of Responsibility

- **Module Lead – Complaint Management System, Fusion (IIITDMJ Campus Platform)** *2025*
  - Architected comprehensive UML and class diagrams for Complaint Management System, defining critical entities and workflows for cross-module integration reference.
  - Spearheaded a 10+ member engineering team, reviewing pull requests weekly against strict coding standards and resolving complex git merge conflicts to ensure seamless cross-module integration before production deployments.

Skills

- **Programming Languages:** C/C++, Python, SQL, JavaScript
- **Computer Science Concepts:** Data Structures & Algorithms, System Design (HLD & LLD), Object-Oriented Design (OOD), Agile Development (Scrum), Software Development Life Cycle, RESTful APIs, Computer Networks (TCP/IP, Routing, WebSockets), Operating Systems
- **Technologies & Tools:** Node.js, React, FastAPI, MongoDB, PostgreSQL, Docker, Redis, Socket.IO, JWT, bcrypt, Cloudinary, Git, Playwright, Linux, LangChain, FAISS, HuggingFace, Streamlit

Certifications

- **Software Testing – NPTEL (Elite, 75/100)** View Certificate
  Applied in automating form submission using Playwright Jul 2025 - Oct 2025
- **Ethical Hacking – NPTEL (90/100)** View Certificate
  Studied web vulnerabilities (SQLi, XSS) to design secure backend systems Jul 2025 - Oct 2025